MEMORANDUM

To:	Strategic Planning Committee JPMorgan Chase & Co.
From:	Iraj Nayebkhil
Date:	April 1, 2025
Subject:	Proposal for Early Adoption and Deployment of Quantum Computing at **JPMorgan Chase & Co.**

Executive Summary

This memorandum presents a compelling strategic case for JPMorgan Chase to adopt and progressively deploy quantum computing technologies as a response to mounting complexity, data intensity, and competitiveness in the financial services industry. As traditional computing systems reach their functional limits, particularly in areas like portfolio optimisation, risk modelling, fraud detection, and algorithmic trading, quantum computing offers a transformative solution capable of delivering speed, accuracy, and scalability far beyond classical systems.

The memo further outlines the strategic imperative, contextualises key global, market, and technological trends, and evaluates five deployment options that vary by complexity, investment level, and risk, allowing the firm to tailor adoption based on organisational readiness and long-term innovation goals.

The strategic planning committee is urged to fast-track this initiative to senior leadership to ensure JPMorgan Chase stays ahead of the technological curve, seizes early-mover advantages and thus, builds long-lasting resilience.

I. Introduction

Quantum Computing is a recent, rapidly emerging technology that uses the methods and laws of quantum mechanics to solve problems that are too complex for traditional computers (or even supercomputers) to solve. Unlike classical computers, which process information in **binary bits** (each bit being either 0 or 1), quantum computers use **quantum bits**, or **qubits** which allow quantum computing to run several processes simultaneously. Furthermore, when qubits are entangled, the state of one qubit is dependent on the state of another, which is key as it allows quantum computers to coordinate information, making them **significantly faster**. One of the key moments in the development of quantum technologies was Google's Sycamore processor, completing "a complicated computation in 200 seconds… would take over 10,000 years for the world's best traditional supercomputers." in 2019 (Tepanyan, 2025)

Traditional methods in the financial industry are increasingly ill-equipped to handle the growing complexity, scale, and speed required in today's markets. It struggles with high-dimensional data, sequential processing, and slow optimization, especially in areas like risk modeling, asset allocation, and fraud detection. As financial operations become more data-intensive and time-sensitive, these limitations pose significant risks. To remain competitive and resilient, there is a pressing need to evolve toward more advanced, efficient, and scalable solutions—precisely where quantum computing offers a transformative path forward. While **classical computing** is good for tasks that are sequential or logically structured, **quantum computing** is particularly effective for solving problems with **complex structures, huge variable spaces, and probabilistic outcomes**.

For a firm like JPMorgan Chase & Co**.**, quantum computing is especially important because the financial industry relies heavily on processing vast amounts of data, complex calculations, and risk assessments — all of which demand both **speed** and **precision**. Quantum computing thus becomes a disruptive technology that offers a fundamentally more powerful way to handle high-stakes operations to better handle complex interdependencies.

II. Strategic Imperative: The Disruptive Opportunity

One of the most promising applications of quantum computing in finance is in **portfolio optimization**. Quantum computers can process vast solution spaces simultaneously using quantum algorithms, identifying optimal portfolios faster and more accurately. This represents a significant opportunity for JPMC to enhance its financial operations. Quantum computing uses quantum parallelism, the ability to evaluate a large number of potential portfolio configurations simultaneously. Instead of testing one combination at a time (as in classical systems), quantum algorithms like the Quantum Approximate Optimization Algorithm (QAOA) or Variational Quantum Eigensolver (VQE) can explore an exponentially larger solution space more efficiently. This leads to faster convergence to optimal or near-optimal solutions, even when dealing with thousands of variables and complex market constraints. For JPMC, this has direct implications on:

- Multi-asset allocation decisions, especially in fast-moving markets
- Scenario-based stress testing of portfolios under shifting macroeconomic conditions.
- Real-time optimization in **high-frequency or algorithmic trading environments**.

Beyond basic allocation, quantum computing enables the development of **digital twins**, dynamic (IBM, n.d.), high-fidelity simulations of JPMC's financial positions. These digital twins can incorporate real-time data and simulate a variety of future macroeconomic and geopolitical scenarios (e.g., inflation shocks, interest rate changes, regulatory shifts). This allows JPMC to model not just best-case returns, but also tail risks, collateral exposures, and liquidity constraints under stress conditions. Using this insight, the firm can **optimize capital allocation** (Gschwendtner, Morgan and Soller, 2023).

Further, effective **risk management** is fundamental to JPMC's ability to navigate the complexities of global financial markets. Risk models such as Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) are essential tools for quantifying potential losses in portfolios and research has demonstrated that quantum algorithms can estimate VaR and CVaR with improved efficiency, enabling more timely and accurate risk assessments (Stamatopoulos et al., 2024). In addition to risk management, quantum computing also holds promise in enhancing **fraud detection and credit scoring**. Traditional machine learning models often struggle to capture complex and subtle patterns in large and diverse datasets. Quantum-enhanced machine learning can process these multidimensional patterns more effectively (IBM, n.d.).

Integrating quantum computing into JPMC's risk management framework offers **enhanced decision-making**, capitalizing on opportunities while mitigating potential threats and **regulatory compliance** (IBM, n.d.) by providing precise risk metrics.

As evident from Table 1, the firm can achieve significant improvements in **efficiency, accuracy, and strategic decision-making**, positioning itself at the forefront of financial innovation with quantum technologies due to its disruptive potential in dealing with the above financial operations at speeds and scales unattainable by classical systems. Thus, It is imperative for senior management to prioritize investment in quantum computing to maintain and enhance the firm's competitive edge in the evolving financial landscape.

Operations JPMC	Traditional Approach	Quantum Computing Approach
Portfolio Optimization	Utilizes **classical algorithms**; computationally intensive with increasing complexity.	Quantum algorithms process **vast solution spaces simultaneously**, identifying optimal portfolios more efficiently.
Risk Assessment (VaR, CVaR)	Relies on extensive simulations; **time-consuming** for large portfolios.	Quantum methods **accelerate risk calculations**, enabling more timely and accurate assessments.
Fraud Detection and Credit Scoring	Uses classical machine learning models; **may struggle with complex data patterns**.	Quantum computing **enhances pattern recognition, improving accuracy** in fraud detection and credit scoring.
High-Frequency Trading (HFT)	Employs classical algorithms; **limited by processing speed** and data analysis capabilities.	Quantum processors enhance **real-time strategy optimization** and market signal extraction.
Regulatory Compliance and Reporting	Involves manual processes and classical computations; **prone to delays and errors.**	Quantum computing **streamlines compliance processes**, ensuring faster and **more accurate reporting**.

Table 1 - Traditional Approach vs. Quantum Computing Approach (IBM, n.d.)

III. Context: Key Trends and Future Change Drivers

Let's understand the key trends and the future change drivers in the finance industry to better understand the opportunities of quantum computing and anticipate disruption.

Global Trends

- *Big Data:* The exponential growth in data volume, velocity, and variety has outpaced the processing capabilities of classical computers (Tepanyan, 2025).
- *Macroeconomic and Geopolitical Volatility:* Quantum computing enables simulation of diverse economic scenarios, allowing firms like JPMC to prepare for interest rate fluctuations (Gschwendtner, Morgan, & Soller, 2023).

Market Trends

- *Rising Complexity in Financial Products:* As financial instruments become more intricate, traditional methods struggle to model and analyze them efficiently, quantum computing manages this complexity (Stamatopoulos et al., 2024).
- *Competitive Innovation Race:* Major banks like Goldman Sachs and Citibank are already exploring quantum computing applications, making it imperative for JPMC to not fall behind (Gschwendtner, Morgan, & Soller, 2023).

Technology Trends

- *Quantum Advantage Achieved:* Demonstrations such as Google's Sycamore processor prove the potential of quantum systems (Tepanyan, 2025).
- *Financially-Oriented Quantum Algorithms:* Tools like the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE) are specifically designed to solve optimization relevant to finance (IBM, n.d.).
- *Quantum Machine Learning (QML):* Emerging QML techniques are redefining fraud detection, credit scoring, and anomaly recognition by identifying complex patterns in massive datasets with high accuracy (Stamatopoulos et al., 2024).

Future Change Drivers

- *Hybrid Quantum-Classical Systems:* Quantum computing's fusion with AI is expected to deliver intricate decision-making capabilities in both speed and precision.
- Recent projections suggest that quantum-computing use cases in the finance industry could generate **$622 billion in value by 2035** (Gschwendtner, Morgan, and Soller, 2023) (Figure 1)

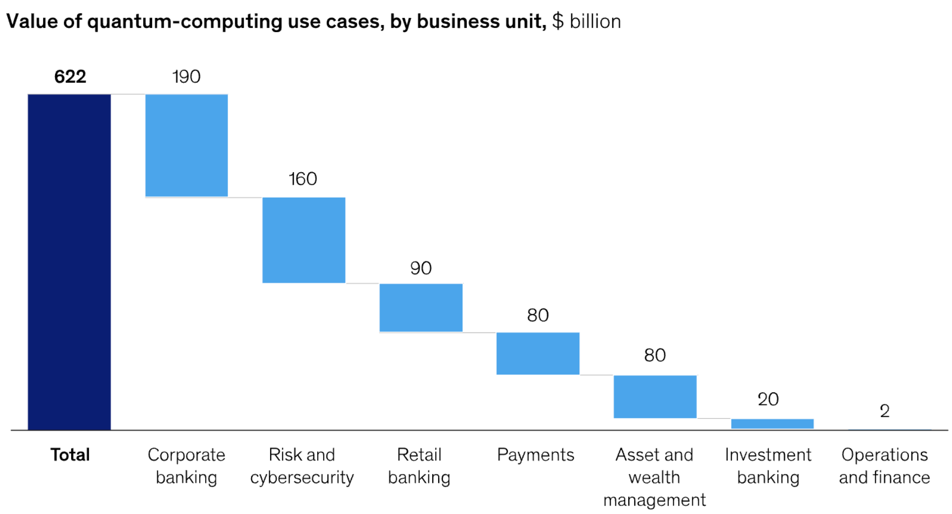

Figure 1 - McKinsey & Company - Use cases in finance by 2035 (Gschwendtner, Morgan and Soller, 2023)

- *Economic Volatility & Uncertainty:* The post-pandemic global economy, inflationary pressures, and market instability increase the demand for scenario modeling and real-time optimization, where quantum computing provides competitive advantage.

Thus, the convergence of global, market, and technology trends, along with powerful future change drivers, **clearly underpins the strategic relevance of quantum computing** due to the exponential **growth of data and rising economic volatility** demand faster, more robust computational tools, while **increasing complexity in financial products** and heightened regulatory pressure expose the limits of classical systems. Moreover, the shift toward hybrid systems and the industry-wide race for innovation emphasize the need for early adoption. For JPMC, these factors make quantum computing not just an opportunity, but a strategic imperative for resilience, competitiveness, and long-term growth.

IV. Deployment Pathways

JPMC must carefully consider its approach to integration, given the complexity of financial operations and the disruptive potential of quantum technologies. A phased deployment approach is essential and elaborated on in the following options, ranging from early adoption methods with low-risk to more advanced, long-term investments.

Option 1 *– Hybrid Quantum-Classical Integration*

JPMC could initially start off by using quantum computing for sub-tasks within larger classical workflows. For instance, using quantum solvers for the hardest parts of a risk model while leveraging classical systems for data ingestion and reporting. Tools like Qiskit, Amazon Braket, and Pennylane support this hybrid architecture. These hybrid systems allow JPMorgan to scale computational efficiency in bottleneck areas while preserving existing systems that management is used to working with. It is best to consider this in the early stages of deployment, as it integrates quantum gradually into familiar workflows.

Option 2 *– Quantum-as-a-Service (QaaS)*

JPMorgan can utilize cloud-based quantum services offered by providers like IBM, Google, and Amazon. These platforms allow real-time access to quantum hardware or simulators for testing algorithms and running proofs of concept. Building quantum infrastructure is indeed expensive and not yet mature for mass deployment. QaaS addresses the need for flexibility and scalability, enabling JPMorgan to run experiments and develop use cases like fraud pattern detection without deep upfront investment in physical systems. It is best to consider this in the early stages of deployment, especially for pilot projects and proof-of-concept developments without committing to capital-heavy projects.

Option 3 *– Partnership with Quantum Technology Providers*

JPMC can deepen collaborations with quantum leaders like IBM, D-Wave, or Rigetti, gaining access to their quantum infrastructure, expertise, and cloud platforms. This approach allows the firm to experiment with quantum algorithms and test early-stage applications without building the hardware or full internal teams. Partnering with quantum providers allows JPMorgan to tap into systems capable of solving these problems faster, especially for use cases like real-time portfolio optimization and risk simulations, which are computationally intensive and slow on classical machines. It is best to consider this in the early-to-mid stages of deployment, as partnerships offer great speed and innovation through shared expertise.

Option 4 *– Strategic M&A of Quantum Startups*

By acquiring startups focused on quantum finance, JPMorgan can integrate innovative technology and talent pipelines into its ecosystem, fast-tracking the adoption of novel solutions. M&A allows JPMorgan to bypass R&D delays, gaining immediate access to specialized capabilities like quantum-enhanced credit scoring or alternative risk prediction models, something traditional models can't efficiently process. This is best considered in the later stages when the firm and its management has warmed up to quantum technologies and a foundational understanding and confidence is established within the organization.

Option 5 *– In-House Quantum R&D and Development*

JPMC could invest in building an internal quantum computing division to create custom algorithms and proprietary solutions tailored to the firm's specific needs. While this approach requires significant finance allocation and management, it allows JPMorgan to directly tackle firm-specific challenges such as tailored risk metrics, custom scenario modeling, and sensitive compliance constraints that require deeper control over modeling and infrastructure, providing a direct competitive edge over firms with classical solutions that are often generic and rigid, lacking flexibility to adapt to complex financial systems. Similar to Option 4, this is best adapted in the later stages when the firm is ready to invest in long-term internal innovation and owns sufficient quantum fluency.

When choosing from the above deployment (*1-2*) and development (*3-5*) options, firms should consider their readiness of the management to adapt, available resources, strategic priorities, and overall risk adversity. **The correct pathway depends on whether the firm demands an initial experimentation, long-term innovation, or gradual integration.** Thus it is **highly probable** that **JPMorgan Chase & Co.** will successfully unlock the full transformative potential of quantum technologies in the financial sector.

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